

Module Plan: Responsible and Trustworthy AI (with focus on Explainable AI, Causal AI, and related concepts like fairness, robustness, privacy, and accountability).

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(Causal part will be removed if already covered)

Key Topics to Include (Brainstorming)

- **Core Principles of Trustworthy/Responsible AI:** Fairness (bias detection/mitigation), transparency, accountability, robustness/safety, privacy (e.g., differential privacy), reliability.

Book to follow while creating content: [Trustworthy AI](#)

- **Explainable AI (XAI):** Black-box vs. interpretable models; post-hoc methods (LIME, SHAP); intrinsic interpretability (decision trees, rule-based, monotonic models); evaluation of explanations (fidelity, stability, human-understandability).
- **Causal AI:** Ladder of causation (association, intervention, counterfactuals); Structural Causal Models (SCMs), Directed Acyclic Graphs (DAGs); DoWhy/Do calculus; moving beyond correlation to actionable insights and robust explanations.
- **Related Terms/Concepts:** Adversarial robustness, model auditing, regulatory frameworks (e.g., EU AI Act risk categories, NIST AI RMF), ethical considerations, human-AI collaboration/trust calibration.
- **Intersections:** How XAI + Causal AI support trustworthiness (e.g., causal explanations are more reliable than purely correlational ones).

Resources to Prepare From

- NIST AI Risk Management Framework (AI RMF).
- Christoph Molnar's *Interpretable Machine Learning* (free online book).
- Papers/books on XAI (e.g., *Explainable AI: Interpreting, Explaining and Visualizing Deep Learning*).
- Causal inference resources (e.g., *Causal Inference for the Brave and True*, DoWhy tutorials).

Ideas for Assignment for the Week

Try First and Understand Later: Learners attempt to explain a black-box model (e.g., a trained XGBoost or neural net on a tabular dataset) using basic feature importance or partial dependence plots before formal XAI/Causal lectures. They reflect on limitations (e.g., “Why might this mislead on causality?”).

Learning Objectives (Bloom's taxonomy; low floor/high ceiling/wide walls):

By the end of this module, all learners will be able to (minimum expectation):

1. **Define and explain** the core principles of Responsible and Trustworthy AI (fairness, transparency, accountability, etc.) and their relevance to real-world deployment (Remember/Understand).
2. **Apply** basic XAI techniques (e.g., SHAP or LIME) to interpret predictions of a pre-trained model and articulate strengths/limitations (Apply/Analyze).
3. **Describe** the differences between correlation and causation, and identify scenarios where causal reasoning improves trustworthiness (Understand/Analyze).

By the end of the lesson, some students will be able to (depth/diversification):

1. **Implement and compare** multiple XAI methods (e.g., SHAP vs. counterfactuals) and **evaluate** them using metrics like fidelity or human-subject studies (Analyze/Evaluate/Create).
2. **Build and query** a simple causal model (e.g., using DoWhy) for intervention or counterfactual analysis on a dataset (Apply/Create).
3. **Critique** an AI system against frameworks like NIST AI RMF or EU AI Act and propose mitigations (Evaluate/Create).

Prerequisites:

- Basic machine learning (supervised models, scikit-learn, basic deep learning).
- Python proficiency (pandas, matplotlib/seaborn, Jupyter).
- Introductory statistics/probability (correlations, p-values).

Problem Set

Context: In high-stakes domains like healthcare, lending, or hiring, opaque AI models can perpetuate biases, lack accountability, and fail under distribution shifts. Regulators and users demand transparency and causal understanding for trust and compliance (e.g., EU AI Act high-risk systems). This module uses techniques from XAI and Causal AI to build more trustworthy systems.

Dataset: UCI Adult Income (Census) or a healthcare/readmission dataset (or COMPAS for fairness discussion) which are common for bias/fairness + XAI examples. Alternative: Synthetic data with known causal structure for Causal AI.

Task:

1. Train a black-box model (e.g., Random Forest/Gradient Boosting or simple NN) for prediction (income >50k, readmission risk, etc.).
2. Apply global and local XAI methods (feature importance, SHAP, LIME) to explain predictions; visualize and interpret.

3. Conduct bias/fairness audit (e.g., by protected attributes like gender/race).
4. (Extension) Build a causal graph, estimate effects, and generate counterfactuals; compare to purely predictive explanations.
5. Reflect on trustworthiness: How do these methods address regulatory/ethical concerns?

Submission Requirements:

- A Jupyter Notebook (.ipynb) containing the complete analysis, code, visualizations, and commentary.
- Clearly comment all code blocks.
- Answer all the required steps explicitly (include a dedicated reflection section on limitations and responsible use).

Online Learning Roadmap — Resource Design Notes

- **Minimum viable resource set:** Free online book chapters + library tutorials (SHAP, DoWhy) + one video.
- **Low-floor entry:** Interactive notebooks with pre-trained models and guided cells; Molnar's book (very accessible).
- **High-ceiling extension:** Implementing from scratch (e.g., custom SHAP approximations), causal discovery algorithms, or regulatory mapping projects.
- **Multiple application pathways:** Tabular, image (e.g., saliency maps), NLP (attention/LIME), or domain-specific (healthcare fairness).

Articles / Tutorials / Book Chapters:

- Christoph Molnar — *Interpretable Machine Learning* (core reference).
- NIST AI RMF documentation.
- DoWhy/EconML tutorials or *Causal Inference for the Brave and True* (Matheus Facure).
- IBM or Red Hat responsible AI overviews.

Videos / Slides:

- "Introduction to Explainable AI" videos (e.g., on SHAP/LIME).
- Causal AI webinars (e.g., TWIML on causal modeling).
- Coursera XAI Specialization (optional deeper dive).

Scaffolded Starter Notebooks:

- SHAP GitHub examples + DoWhy tutorials (pre-loaded dataset + model).
- Kaggle XAI notebooks.

AI Prompts for Guided Exploration:

- "Explain the SHAP values for this prediction in simple terms for a non-technical stakeholder."

- “Generate counterfactual examples for this model input and interpret causally.”

Mid-Week Hints / Guardrails:

- Common pitfalls (e.g., SHAP assumptions, confounding in causal models).
- When to prefer intrinsic vs. post-hoc explanations.

Discussion Prompts for Peer Debugging:

- “Share a case where your XAI explanation surprised you or seemed misleading : why?”
- “How might causal insights change deployment decisions compared to correlational ones?”

Session Title [Estimated Time]

Session 1: Foundations of Responsible & Trustworthy AI (60-90 min)

Session 2: Explainable AI Techniques (90-120 min)

Session 3: Causal AI for Deeper Trust (90-120 min) : *This is supposed to be done in previous classes*

Session 4: Integration, Auditing & Regulation (60-90 min + project work)

Brainstorming:

- **Focus on Cognition, not Coding:** Critical thinking on when explanations are trustworthy vs. performative; distinguishing correlation from causation; evaluating explanations for human usefulness and potential misuse.
- **Learning to Learn:** Reflection habits (e.g., “Did this explanation build or erode trust?”); tool-agnostic reasoning (understand *why* a method works beyond library calls); adapting to new regulations/domains.

Introductory Motivation:

AI systems increasingly impact lives (hiring, loans, medicine), but black-box nature erodes trust, amplifies biases, and hinders accountability. Regulations like the EU AI Act and frameworks like NIST demand transparency and risk management. XAI and Causal AI provide tools to make AI understandable, fairer, and more reliable.

Topics:

As listed in Key Topics.

Tests for Chapter-Level Knowledge/Skills:

- Quiz on principles and differences (XAI vs. Causal).
- Interpret a provided SHAP plot or causal graph.
- Short essay: “How would you explain this model decision to a regulator?”

Teaching Methodology:

- **ReCap:** Quick poll on prior experiences with “unfair” or opaque AI.
- **Plenary:** Principles + frameworks discussion; live demos of SHAP/Causal queries.
- **Demo:** Walkthrough notebook (black-box → explained → causal).
- **PreCap:** Tease next session’s extension (e.g., counterfactuals for “what if” fairness interventions).
- **References:** NIST, Molnar book, DoWhy docs, key papers.

Main Takeaway:

Responsible AI isn’t just about accuracy, it’s about building systems that are transparent, causally grounded, fair, and accountable so humans can trust and effectively oversee them. XAI and Causal AI are practical enablers for this.

For Internal KSS (For AI / Software Engineers)

Mainly Prompt and Agentic Focused

1. Preventing Prompt injection (Using the existing libraries and Manual Guardrails creation)
2. Vulnerability Identification
3. Data Privacy (what to provide? What not to provide ? For governance)
4. What is and How to Handle PII?
5. Handling Sensitive information
6. Input output validation (in AI / LLMs)
7. Biasedness detection
8. Alignment Issues (Look up Anthropic for this)
 - a. Look to alignment for all the resources
 - b. How is Anthropic doing it?
9. Handling the ->Safe request refusal?
10. Hallucinations
11. Alignment Issues & Autonomy Control (Human in/on/out the loop) (Psychofancy)
12. Governance
13. Data Poisoning
14. Incident Management - (If Losses occur , who bears the cost?)